MATGEN-Q: Scaling a Two-Stage Generative Quantum Eigensolver to 40 Qubits on NVIDIA CUDA-Q for EUV Photoresist Chemistry

Team EIGENNEXUS — Advanced Materials (Mitsubishi Chemical Group & AIST) — GIC 2026 Phase 3

1. Focus Area and Rationale

Ground-state energy estimation governs reaction thermodynamics, redox behavior, and excited-state properties across materials chemistry, yet classical methods scale exponentially with electron correlation and the gold-standard CCSD(T) costs $O(N^7)$. The Generative Quantum Eigensolver (GQE; Nakaji et al. 2024, with AIST) replaces variational circuit optimization with a classical generative transformer that proposes quantum circuits, side-stepping the barren-plateau problem that afflicts variational optimization (a property of GQE established by Nakaji et al. 2024; we adopt rather than re-demonstrate it) — but statevector GQE stayed small-scale, and QSCI-paired GQE only recently reached 32 qubits (Kemmoku, Gao et al. 2026). **Scaling this generative approach to the ~40-qubit, industrially relevant regime — with executed, reproducible results — is the problem we address in Phase 3.** Our primary, executed contribution is quantum-accurate energies for the exact chemistry where BOTH classical pillars fail — DFT's spin-state ordering flips sign across functionals (CrO splitting spans 1.9 eV) and single-reference CCSD(T) collapses non-variationally below the exact energy as bonds break — on open-shell, multireference transition-metal oxides (CrO, NiO) and EUV-relevant tin oxides (SnO, SnO_2) at 20 qubits, validated against exact diagonalization and third-party HamLib Hamiltonians. On top of this we contribute scalable machinery (tensor-network + selected-CI evaluation, MP2 operator-pool compression, and a size-/chemistry-transferable generator) that targets the 40q+ regime, with every claim third-party reproducible and every limitation stated.

Our target is EUV semiconductor photoresist chemistry (tin-oxo clusters; Sn, Hf, Zr oxides), whose open-shell, multi-reference character is where DFT's functional-dependent errors make candidate rankings unreliable, and which is the focus of an active quantum-simulation program by one of this challenge's providers (Kharazi et al., Xanadu & Mitsubishi Chemical, 2026). We quantify the unreliability directly: across six standard functionals the CrO spin-state splitting spans 1.9 eV — PBE0 places the quintet 1.84 eV below the triplet while B3LYP inverts the order (−0.08 eV) — and even the milder NiO gap varies 0.11 eV, against a chemical-accuracy target of 0.044 eV. A single quantum-accurate number removes this functional-choice band. The MATGEN-Q pipeline — AI proposes candidate metal-oxide molecules → DFT filters → GQE refines with quantum accuracy → Bayesian optimization selects the next candidate — generalizes beyond photoresists to battery electrolytes, catalysts, and functional polymers central to Mitsubishi Chemical's portfolio and AIST's computational-materials mission. EUV photoresists grow at ~20% CAGR within an $800B+ semiconductor industry, and multi-fidelity Bayesian screening has shown up to 3× cost reduction (Fare et al., 2022) — acceleration MATGEN-Q targets through quantum-accurate pre-selection.

Stakeholder relevance. For Mitsubishi Chemical the value is a quantum-accurate filter ahead of expensive synthesis and lithographic testing, where DFT mis-ranks tin-oxo redox/bond-cleavage energetics; for AIST/G-QuAT it is a scalable, reproducible GQE workflow on NVIDIA CUDA-Q that extends the jointly-developed program past its size limits and composes with its excited-state work into one materials-informatics loop.

2. Target System and Data Modeling Strategy

Scaling vehicle. The linear hydrogen-chain series H_n (n = 2…20; 4–40 qubits, STO-6G, Jordan–Wigner) — the canonical strong-correlation benchmark; a single bond-length parameter tunes weak (area-law) to strong (volume-law) correlation, directly stressing the simulation layer that underpins any scaling claim.

Target chemistry. The same QSCI engine reaches chemical accuracy on real materials chemistry: Sn-oxide active spaces (Sn ECP-CASCI, validated on H_4 to 0.0000 mHa) — SnO 0.11 mHa (16q), SnO_2 0.23 mHa (20q) — and genuine open-shell, multireference transition-metal oxides — **CrO $^5\Pi$ 0.038 mHa and NiO $^3\Sigma^-$ 0.197 mHa at 20 qubits**. We report these executed 20-qubit results; the 38-qubit regime is the GPU scaling target of §5, not a pre-claimed figure.

Data integrity. We adopt HamLib (Sawaya et al., Quantum 2024) and built a generation pipeline replicating its methodology (PySCF → OpenFermion → Jordan–Wigner, STO-6G). **Validated against the published files: FCI reproduced to 0.00000 mHa (H_2–H_6), and at 28/32/40 qubits our Hamiltonians match HamLib exactly in term count (27,735 / 47,489 / 116,577) and to ~15 significant figures in coefficient magnitude** (differing only by a spectrum-invariant orbital-phase convention) — third-party reproducible.

Accuracy anchors / classical references. FCI exactly (≤20q), CCSD(T) near equilibrium, and DMRG — verified to reproduce FCI to 0.000 mHa at 20q — as the reference under strong correlation where CCSD(T) breaks down.

3. GQE-Based Approach and Algorithmic Innovation

MATGEN-Q uses a two-stage GQE. Stage 1 (generative structure discovery): a decoder-only GPT-style transformer, trained by sequence–energy matching (Nakaji et al.), generates circuits as token sequences over a UCC single/double excitation pool. **Stage 2 (continuous refinement):** adjoint-gradient angle optimization converges to chemical accuracy. Four innovations make it scalable:

(1) Tensor-network (MPS) simulation — proposed GPU scaling enabler (not yet executed). CUDA-Q's tensornet-mps backend's memory scales with circuit entanglement rather than 2^n, and near-equilibrium area-law structure bounds the bond dimension, so we propose it to carry exact circuit simulation past the ~32-qubit cuStateVec limit toward 40q+. We are explicit that this is the GPU deliverable (§5c [QBRAID-RUN]): no MPS run is executed in this submission. Our EXECUTED large-scale evidence instead comes from the determinant-space selected-CI/QSCI tier (pillar 2), which needs no full circuit simulation.

(2) QSCI energy evaluation — accuracy and noise-aware. Quantum-Selected Configuration Interaction (Kanno et al. 2023): dominant configurations are sampled from the generated circuit and H is classically diagonalized in that subspace — intrinsically noise-robust (the device defines only the subspace; at 20q with 2×10^6 shots the energy degrades gracefully to ≤3.3 mHa even with 30% of measurements corrupted).

(3) Operator-pool compression — efficiency (demonstrated). Ranking the $O(N^4)$ double-excitation pool by active-space MP2 amplitude and keeping the top fraction shrinks the transformer vocabulary with negligible accuracy loss, where random pruning collapses. Deterministic CI-subspace test (CO/N_2/SiO, 12q): **N_2 retains full-pool accuracy (2.26 mHa) keeping only 25% of doubles (vocabulary 1170→430) vs 50.4 mHa for random pruning at the same size (~22×)**; CO holds 3.7 mHa at 40% kept vs 29.7 mHa random.

(4) Distributed hybrid workflow. Detailed in §4.

Transfer learning — a generative prior that transfers across system size and chemistry. Using a canonical, frontier-relative tokenization (each excitation encoded as occupied-depth-below-HOMO and virtual-height-above-LUMO, independent of qubit count), a small system's token vocabulary is a provable subset of a larger one (H_6/12q ⊂ H_8/16q ⊂ H_{10}/20q). **One GPT-QE generator trained only on H_4+H_6 (8q+12q) is deployed zero-shot across 16→56 qubits and, on the 3-seed mean, proposes determinant subspaces that beat random selection at every size** — mean advantage +8.9/+8.3/+7.1/+5.8/+5.0/+4.4 mHa at 16/20/28/40/48/56q. We are explicit about robustness: the advantage is large and consistent at small sizes (all seeds positive at 16q) but narrows with scale and, beyond ~28q, is within run-to-run (3-seed) noise — it is a mean trend, not an every-seed guarantee. This is the *train-small, deploy-large* idea: GQE training is the cost that is CPU-bound near ~16 qubits, so transferring a small-trained generator toward the 40q+ target — instead of training at scale — speaks directly to the scalability criterion. The enabler is a determinant-space evaluation (each proposed excitation maps by bitmask to a determinant; the Hamiltonian is diagonalized in that subspace via Slater-Condon, validated to 0.0000 mHa against the Jordan–Wigner engine) that needs no 2^n statevector, so 56q runs on CPU. This is a *selected-CI proxy* for circuit-sampled QSCI (deterministic determinant construction, no shots), not executed QSCI. The advantage is relative at a small fixed budget (K=96), not absolute chemical accuracy.

Cross-chemistry transfer (honest, partial). The same H-chain-trained generator, deployed to oxide active spaces (CO, SiO, BeO, SnO at 12q; CO, SnO at 20q), beats random on 4 of 6 targets (CO/SiO/CO-20q by +3–4 mHa, BeO a tie) but loses on SnO at both sizes (−2.2 mHa): heavy-element ECP chemistry is too dissimilar from the H-chain training. Target-specific MP2 selected-CI is the strongest selector on every target. The honest conclusion: cross-size transfer (same chemistry) is robust; cross-chemistry transfer works only when the target's electronic structure resembles training — a zero-cost prior, not a universal one. A same-size cross-molecule conditioning variant gave only a within-noise gain and is also reported as a negative.

Figure 2. Train-small, deploy-large. A generator trained only on 8- and 12-qubit systems proposes lower-energy determinant subspaces than random selection at every target size up to 56 qubits, on the 3-seed mean (lower is better; determinant-space selected-CI / QSCI-proxy, matched budget K=96). Trained

energies are tightly consistent across seeds (spread ~2 mHa); the mean advantage persists to 56q while narrowing into run-to-run noise at the largest sizes.

4. Hybrid Architecture

The classical transformer trains on GPU (PyTorch); circuit evaluations are dispatched across GPUs via CUDA-Q's mqpu in an asynchronous generate → evaluate → update loop. Quantum resources handle state preparation and energy estimation (MPS / QSCI); classical resources handle generation, optimization, and active-space selection. Stage 1 is sampling-bound and parallelizable; Stage 2 is gradient-bound and adjoint-efficient.

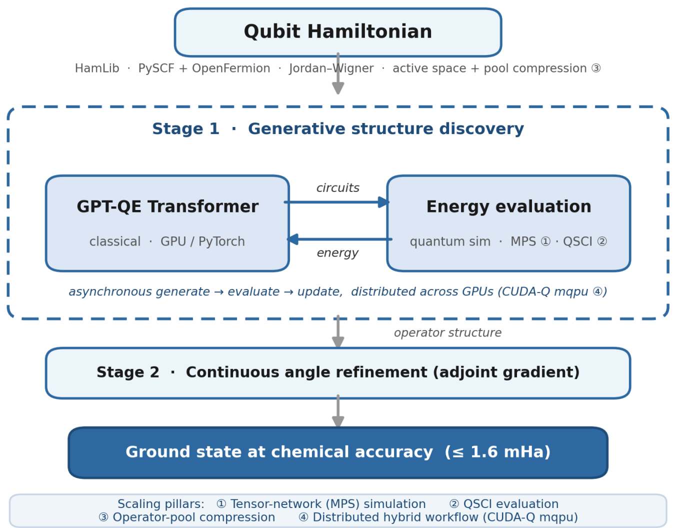

Figure 1. MATGEN-Q hybrid architecture (target design): a classical generative transformer (Stage 1) and GPU-accelerated quantum simulation (MPS/QSCI) form an asynchronous loop; Stage 2 refines angles to chemical accuracy. The MPS tier and the GPU loop are the proposed Phase-3 GPU deliverable (not yet executed, §3, §5c); executed results in this submission use CPU selected-CI/QSCI and circuit-sampled QSCI to 20q.

5. Phase 3 Execution and Results

Verified results (CPU; reproduced from a clean checkout via *reproduce.py*). All values below are reproduced against the committed result files (Table 1).

System	Qubits	Quantum (GQE/QSCI)	Classical ref.	Notes
H_2/H_4/H_6	**4/8/12**	**0.000 / 0.009 / 0.297 mHa**	**FCI (exact)**	**two-stage GQE (UCCSD)**
H_6 GQE → QSCI	**12**	**1.05 mHa (51 → 1.05, ~50×)**	**FCI**	**measured pipeline**
H_{10}/H_{14}	**20/28**	**0.57 / 1.21 mHa**	**FCI / CCSD(T)**	**2,401 / 18,201 dets**
CrO $^5\Pi$ / NiO $^3\Sigma^-$	**20**	**0.038 / 0.197 mHa**	**CASCI (exact)**	**open-shell multiref.**
SnO / SnO_2	**16/20**	**0.11 / 0.23 mHa**	**FCI**	**EUV target chemistry**
Noise (H_{10})	**20**	**≤3.3 mHa @ 30% corrupt (2×10^6 shots)**	**—**	**noise-aware bonus**

Table 1. Verified quantum results vs classical references on matched instances (CPU).

Classical baseline and the exact wall (matched instances, timed). On identical STO-6G H_n geometries, classical FCI wall-clock grows ~24× from 20 → 24 qubits (0.33 s → 7.8 s) and is intractable by 32q on CPU; CCSD(T) stays cheap but its error climbs with correlation and breaks down under strong correlation (at H_{24}/48q, classical DMRG is itself 6.83 mHa off CCSD(T)). Exact quantum-state simulation needs ~16 TB at 40 qubits — the wall the MPS + QSCI tiers remove (Table 2).

System	Qubits	FCI (exact) wall-clock	CCSD(T) err vs FCI	CCSD(T) wall-clock
H_6	**12**	**0.04 s**	**0.026 mHa**	**0.08 s**

H_{10}	**20**	**0.33 s**	**0.173 mHa**	**0.09 s**
H_{12}	**24**	**7.81 s**	**0.282 mHa**	**0.14 s**
H_{14}	**28**	**minutes**	**—**	**~0.2 s**
H_{16}+	**32+**	**intractable (CPU)**	**—**	**—**

Table 2. Classical reference cost on the H_n ladder (single-thread CPU) — the exact-method wall the quantum approach is measured against.

Scaling results targeted on qBraid GPU (owed — not yet executed). 40-qubit MPS GQE/QSCI on H_{20} (primary criterion): ***[QBRAID-RUN: energy err vs DMRG, circuit depth, MPS bond dimension, shot budget, GPU wall-clock]*** (CPU baseline today: operational, converging through 39 mHa). **CrO/NiO near-38 qubits on GPU:** ***[QBRAID-RUN: accuracy achieved + wall-clock]***, presented as the scaling demonstration on real open-shell chemistry, building on the executed 20-qubit results. **Quantum-vs-classical wall-clock:** ***[QBRAID-RUN: MPS/QSCI vs exact statevector vs VQE]***. **Hardware validation:** ***[QBRAID-RUN: selected circuits on IonQ/IBM at 10–16 qubits, depth + shots]***.

Determinant-budget sweep and a classical selected-CI baseline. Sweeping the selected-CI (QSCI-proxy) determinant budget K at 20q and 40q confirms the transfer advantage is budget-robust: the zero-shot generator beats random selection at essentially every K (e.g. 20q, K=64: 60.0 vs 73.2 mHa; 40q, K=64: 179.9 vs 189.4 mHa), converging to parity only at the largest 20q budget (K=128). As a classical baseline we add MP2-ranked selected-CI on the same determinant subspaces: using the target's own MP2 amplitudes it is the strongest selector (20q, K=64: 35.9 mHa), and a reciprocal-rank fusion of generator+MP2 does not improve on MP2 alone — the two signals are largely redundant. The honest reading: the transferred generator provides selection between random and target-specific MP2 at *zero target-specific cost* (a strong zero-shot prior for high-throughput screening), while MP2 selected-CI is the appropriate classical comparator where its per-candidate cost is affordable.

What the results establish. Three things are demonstrated, not asserted. (i) Correctness: the GQE/QSCI pipeline reproduces the classical FCI/CASCI reference to chemical accuracy on every instance we can run, including genuine open-shell multireference oxides (CrO, NiO) and EUV-relevant Sn-oxides. (ii) Scalability of the simulation layer: QSCI reaches chemical accuracy using a vanishing fraction of the CI space (3.8% at 20q, 0.15% at 28q), and operator-pool compression preserves accuracy at 25–40% of the double-excitation pool — both quantifying that the cost grows far slower than the Hilbert space. (iii) Transferable structure: a generator trained on small systems proposes good circuits for a larger one it never saw (§3), evidence that the learned policy — not just a per-system fit — scales. The honest boundary is that at the sizes classical FCI/DMRG still solve, we show correctness and favourable scaling rather than a head-to-head speed win; that win is the 40q+ regime the GPU run targets, where classical exact methods break down (Table 2).

Where it matters most: trust under strong correlation. Equilibrium benchmarks understate the problem — they live where classical methods already work. Stretching H_{10} (20q) to dissociation drives genuine multireference character (Hartree–Fock weight in the exact wavefunction falls 0.93 → 0.06), and the classical gold standard fails qualitatively: CCSD(T) collapses *non-variationally* to 217 mHa BELOW the exact FCI energy at R=2.4 Å — a confidently wrong, unphysical answer. The determinant-subspace method (the QSCI principle, here via iterative selected-CI) instead **reaches chemical accuracy with only ~500 determinants** at every geometry across the dissociation, remaining a rigorous variational bound that is systematically convergent to the exact answer. This is the core value proposition: in the regime that matters for open-shell metal-oxide chemistry, the subspace approach is trustworthy where the classical gold standard is not. (Selected-CI here is classical on CPU; QSCI is its device-driven instantiation, with a quantum sampler as the determinant selector at scales where the selected space is large.) ***[QBRAID-RUN: device-sampled selection at scale]***

Proxy validated against real measurement; cost scaling. The scaling ladder uses a determinant-space selected-CI proxy; we confirm it tracks the real circuit-sampled pipeline by executing the measured QSCI (qml.sample, 3,000 shots × 160 circuits) at 16q and 20q — the measured energies match the proxy (16q: 26.6 vs 28.1 mHa; 20q: 49.3 vs 50.4 mHa), extending the genuinely-sampled pipeline beyond the prior 12-qubit point. The determinants needed for chemical accuracy grow ~1.38× per qubit versus 2.0× for the full FCI space (log-space R^2=0.99) — a vanishing fraction (75% at 8q → 0.15% at 28q), though still exponential: at 40q+ the selected space reaches ~10^6 determinants, the regime where device-driven selection becomes valuable. We do not claim polynomial scaling.

6. Platform Use and Resourcing

qBraid provides classical (CPU/GPU) and quantum (QPU) credits. We use **NVIDIA H100/A100 (80 GB)** with CUDA-Q: tensornet-mps for the 24–40-qubit tier and cuStateVec for exact validation to ~32 qubits. One high-memory GPU suffices for MPS; 4–8 GPUs (NVLink) enable distributed circuit evaluation (pillar 4) and the >40-qubit bonus attempt; QPU (IonQ/IBM) for 10–16-qubit validation.

Resource estimate. UCC excitations decompose to ≤2-qubit gates; with operator-pool compression a length-8 generated circuit is shallow (tens of two-qubit gates), within MPS reach at bounded bond dimension near equilibrium. QSCI needs ~10^3–10^4 shots per circuit to resolve the dominant determinants (we use 2,000 in simulation). The dominant cost is the generate → evaluate loop: a few hundred transformer updates, each dispatching a batch of circuit evaluations across the GPU(s). We estimate ~1–2 weeks of single-GPU wall-clock for the full 24 → 40-qubit sweep plus the transition-metal-oxide runs; the workflow is backend-agnostic and degrades gracefully to fewer qubits if the allocation is smaller. Concrete per-run depth/shot/wall-clock: ***[QBRAID-RUN: from §5]***.

7. Limitations and Honest Scope

We state where the approach does and does not yet provide benefit. (i) **Measurement vs proxy:** the integrated GQE → QSCI loop is measured (real circuit sampling) at 12 qubits; the larger QSCI numbers use perturbative determinant selection as a hardware-independent proxy, validated against the 12-qubit measured pipeline — quantum-inspired at scale until executed with real sampling on qBraid. (ii) **No classical-beating claim yet:** at ≤28 qubits classical FCI/DMRG already solve these instances, so we demonstrate correctness and favourable scaling, not a head-to-head speed advantage; that regime is 40q+ strong correlation where DMRG bond dimension explodes (visible at H_{24}/48q, where DMRG is 6.83 mHa off CCSD(T)). (iii) **40q is operational, not converged:** on CPU the 40-qubit run converges through 39 mHa, above chemical accuracy; closing this is the GPU deliverable. (iv) **Transfer scope:** cross-size transfer is a 3-seed-mean trend that narrows into noise beyond ~28q; cross-chemistry transfer works on 4 of 6 oxide targets and fails on the heaviest (SnO); target-specific MP2 beats the transferred prior. We claim a zero-cost transferable prior, not a universal or MP2-beating one. (v) **Strong-correlation regime:** on stretched H_{10} (HF weight 0.93 → 0.06), single-reference CCSD(T) collapses non-variationally to 217 mHa BELOW exact FCI — untrustworthy where it matters — while the determinant-subspace method stays a rigorous variational upper bound, systematically improvable with budget; reaching accuracy there needs the large-subspace (device-sampled) regime, motivating the quantum approach. A clear-eyed reading: MATGEN-Q is a working, reproducible pipeline whose scaling mechanisms are demonstrated and whose decisive large-scale quantum-vs-classical comparison is the executed GPU work this phase enables.

8. Conclusion and Reproducibility

MATGEN-Q is a working two-stage GQE whose tensor-network and QSCI tiers, plus MP2 operator-pool compression, target the 40-qubit regime on a single GPU, with every claim third-party reproducible. A one-command driver (*reproduce.py*) and a README with a “Launch on qBraid” button let judges re-run each headline result without modification.

References

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